

NEHA DHAGE

San Ramon, CA | 610-554-4876 | nehadhage@gmail.com | linkedin.com/in/nehadhage

Recent graduate from School of Information at UC Berkeley with Master of Information and Cyber Security (MICS) and Certificate in Applied Data Science, with academic research and projects focused on AI security, Adversarial Machine Learning, and Policy-driven Governance of emerging technologies. Cybersecurity and Technology risk professional with over nine years of experience in enterprise security governance, risk and control and auditing, across financial services and technology industries. Experienced in translating complex technical risks into actionable governance and control frameworks and collaborating with cross-functional stakeholders including engineers, security teams, regulators, and policy practitioners.

TECHNICAL AND POLICY EXPERTISE

- AI and Cyber Security, Machine Learning (ML) and Adversarial ML, Natural Language Processing and Deep Learning
- AI Risk Governance & Technology Policy, Enterprise Security Governance, Risk and Controls, Risk Assessment and Mitigation.
- Cross-functional Collaboration, Stakeholder/Client Relationship Management and Communication.
- Collaborative documentation tools, Technical Documentation and Analytical Reporting, Policy and procedure creation.

EDUCATION, RESEARCH AND TECHNICAL PROJECTS

University of California, Berkeley - MASTER OF INFORMATION AND CYBERSECURITY (MICS)
CERTIFICATE OF APPLIED DATA SCIENCE

AUG 2024 – MAY 2026
JAN 2025 – AUG 2025

- **Coursework –**
 - Web Application Security Assessment
 - Cryptography for Cyber and Network Security
 - Operating Systems Security
 - Network Security
 - Software Security
 - Security Operations
 - Privacy Engineering
 - Applied Machine Learning for Cybersecurity
 - Natural Language Processing with Deep Learning
 - Research Design and Applications for Data and Analysis
- **Research and Technical projects in Adversarial Machine Learning, Deep Learning, Natural Language Processing and Agent Security-**

1) Agent Credential Firewall (CAPSTONE PROJECT - SPRING 2026)

Agent Credential Firewall is a capstone project that explores how organizations can securely manage and control the use of credentials by autonomous AI agents interacting with enterprise systems. As AI agents increasingly perform tasks such as sending emails, querying databases, and interacting with SaaS applications on behalf of users, they operate using delegated credentials that were originally designed for deterministic software and/or human users. This creates new security risks, including credential misuse, unauthorized data access, automated data exfiltration and confused-agent risk. This project implemented a policy-based enforcement layer, referred to as the Agent Credential Firewall, that evaluates agent actions in real time before they are executed. Using Open Policy Agent (OPA) and a set of defined security rules, the system analyzed contextual signals such as recipient domains, data sensitivity, behavioral patterns, and authorization constraints to determine whether an action should be allowed or denied. Through simulated scenarios and test cases, the project demonstrated how a credential firewall can mitigate risks introduced by agentic systems and provide a practical framework for securing AI-driven automation in enterprise environments. The designed structured scenarios and policy rule sets can serve as instructional case studies for teaching AI security governance and enterprise risk management.

2) Membership Inference attacks and Differential Privacy (FALL 2025)

This project built and evaluated a machine learning models to test how vulnerable it is to membership inference attacks and whether differential privacy can meaningfully reduce that risk. First, convolutional neural network classifiers were trained on standard image classification datasets and a confidence-based membership inference attack was executed. The attack attempted to determine whether a given record was part of the training dataset by using maximum SoftMax confidence as the signal for prediction. Then the models were retrained using Differentially Private Stochastic Gradient Descent (DP-SGD) with varying privacy budgets and re-ran the same attack to measure whether the privacy mechanism reduced the attacker's ability to distinguish members from non-members. Our results showed that non-private models leaked statistically meaningful membership information, while models trained with DP-SGD, especially with stronger privacy parameters, reduced attack success to near-random guessing, albeit with an associated loss in model accuracy.

3) Manipulative Language Detection in LLM-Crafted Phishing Attacks (SUMMER 2025)

This project investigated whether Natural language Processing (NLP) models can effectively detect manipulative language in text, with a specific focus on phishing emails crafted using large language models (LLMs). Since phishing attacks exploit psychological vulnerabilities through subtle, persuasive language, the project focused on identifying manipulation i.e., language intended to influence a reader against their best interest. Using the MentalManip dataset, which contained thousands of film dialogues labeled with fine-grained manipulation techniques, we fine-tuned transformer-based models including BERT and RoBERTa to detect these linguistic patterns. Then, we evaluated whether the models could generalize to real-world phishing emails, especially those missed by existing filters. The research examined whether linguistic manipulation detection could strengthen phishing defenses in cybersecurity context beyond traditional spam detection approaches, and the results showed both the promise and the limitations of this method: while fine-tuned models performed well on manipulation detection, generalization to real phishing emails remained challenging due to class imbalance and format differences between training and target dataset.

4) Fast Gradient Sign Method (FGSM) Adversarial Attack on ResNet-18 model using ImageNet data and defense techniques Gaussian noise and JPEG compression. (SPRING 2025)

In this project, we evaluated the adversarial robustness of a pre-trained ResNet-18 model using samples from the ImageNet dataset. We implemented the Fast Gradient Sign Method (FGSM) to generate adversarial examples by adding carefully crafted perturbations to the input images, aiming to reduce the model's classification accuracy. To counteract these attacks, we tested two defense techniques: Gaussian noise and JPEG compression. Gaussian noise was added as a random perturbation to obscure adversarial patterns, while JPEG compression aimed to remove high-frequency components introduced by the attack. We then assessed the model's accuracy across clean, adversarial, and defended inputs to determine the effectiveness of each defense strategy in mitigating the impact of FGSM. This experiment demonstrated the vulnerability of deep learning models to simple gradient-based attacks and explored the potential of input-level transformations as lightweight adversarial defenses. The project concluded that ResNet-18 is highly sensitive to carefully crafted adversarial perturbations, attack success increases with increase in perturbation strengths, and simple input-transformation defenses can help, with JPEG compression outperforming Gaussian noise for subtle FGSM attacks.

CITRIS and The Banatao Institute – UC Berkeley - TECH POLICY DESIGN: PLATFORM GOVERNANCE FOR AN AI-DRIVEN WORLD (Oct 2025 – Feb 2026)

- This course engaged students from various schools at UC Berkeley with pressing issues in trust and safety, content moderation, algorithmic transparency, legislative process in California and US and emerging regulatory trends by applying design-thinking and policy frameworks to real-world challenges. The course

was led by Camille Crittenden (Executive Director, CITRIS), David Evan Harris, (Chancellor’s Scholar – Haas School of Business), Naomi Schiffman (Oversight Board) and Nichole Rocha (Omidyar Group).

- Analyzed current legislation, amendments, Opposition and Support artefacts, recent technological advances and studies and relevant legislative efforts in other US states.
- Developed concrete policy recommendations for legislations in progress and developed policy recommendations addressing trust, safety, and algorithmic transparency.
- Collaborated with interdisciplinary (Law, Public Policy, Computer and Data Science and Social Science) teams to analyze technology governance challenges and developed presentations and policy analyses for academic and policy audiences.
- Program culminated in final presentations to legislative staffers, policymakers and other key stakeholders at Sacramento in Feb 2026.

CERTIFICATIONS

Cybersecurity Audit Certificate (CSXA) – (#2024-920143-CSXA)	Apr. 2024
Certified in Cybersecurity (CC) – (#729604)	Dec. 2023
Certified Third-Party Risk Assessor (CTPRA) –#45571	Sep. 2021
Certified Data Privacy Solutions Engineer (CDPSE) – #2114689	Mar. 2021
Certified Information Systems Auditor (CISA) – #16135170	Dec. 2016

PROFESSIONAL EXPERIENCE

ISC2 (International Information System Security Certification Consortium). San Ramon, CA
Security Certifications Designer and Review SME **Apr 2024 – May 2024**

- Selected by the International Information System Security Certification Consortium (ISC2) to develop and review globally recognized security certification exams.
- Drafted exam questions and answers, reviewed peer submissions for accuracy and quality, and ensured alignment with industry standards.
- Led and participated in roundtable discussions to shape exam content and maintain the integrity of certification programs for candidates worldwide.

Career Break San Ramon, CA
 Period of focused preparation for Graduate school and Security certifications. **Sept 2023 – Aug 2024**

BANK OF MONTREAL (Formerly Bank of the West). San Ramon, CA
Senior Auditor II, Vice President – Technology Internal Audit **Oct 2017 – Sept 2023**

Led and managed enterprise-wide IT and Cyber security audits with a team of four associates and cross functional teams.

- Performed project management activities and managed the audit plan including timeline, budget, resource availability, communication and relationship with auditees by building collaborative partnerships with auditees and senior executives.
- Presented Internal Audit documentation to Federal regulators periodically and managed communication with regulator resulting in successful Regulatory reviews were successful and without incident.
- Enhanced audit work programs to ensure compliance with standardized audit guidance, and industry best practices, to create customized testing procedures. This helped to build efficiency within Internal audit team.
- Reviewed audit documentation prepared by staff/team to ensure completeness and accuracy of deliverables.
- Mentored and trained junior auditors on cybersecurity frameworks and audit methodology, translating technical concepts into structured training and guidance, resulting in highly trained staff.

ERNST & YOUNG (EY),
Senior IT Auditor — Risk Advisory Services

Sacramento, CA
Jun 2014 – Aug 2017

Led IT advisory and attestation projects in accordance with industry audit standards and regulatory requirements for large public and private entities (more than 10,000 employees) in Technology, Healthcare and Retail industry.

- Assessed design and operating effectiveness of IT and application controls, identified key risks and exceptions in controls framework and provided independent opinions on internal control over financial reporting.
- Issued detailed findings, observations and recommendations to clients.
- Issued SOC1/SOC2/SOC3 (SSAE) reports as per SSAE guidance for public companies in the Technology industry.
- Successfully managed multiple concurrent audit projects while working through substantial scope changes and complexity leading to efficient and timely delivery of audit results.
- Produced structured, technical and written and visual materials explaining complex security systems, risks, and governance frameworks for diverse audiences including regulators, engineers, executives, and policymakers.

SAPIENS (Formerly Adaptik Corporation),
Software Quality Assurance Analyst

Bethlehem, PA
Aug 2012 – Jun 2014

Managed software and system testing, lifecycle of defects, and response to production issues as the Software Quality Assurance Analyst for Property and Casualty (P&C) Insurance software solution suite.

COGNIZANT TECHNOLOGY SOLUTIONS,
Software Engineer

Pune, India
Nov 2009 – Jul 2010

Managed product delivery for Mainframe banking application and acted as liaison between client, offshore, and onsite teams for solution delivery

TECH MAHINDRA (Formerly Satyam Computer Services),
Software Engineer

Pune, India
Sep 2006 – Nov 2009

Designed and implemented new solutions for Mainframe banking applications. Led the entire lifecycle of product development including delivery and support post implementation. Trained new staff on best practices for software development.

PUBLICATION

Published research – Braunschweig, B., **Dhage, N.**, Viera, M. J., Seaman, C., Sampath, S., & Koru, A. G. (2012). Studying volatility predictors in open source software. In *Proceedings of the 2012 ACM-IEEE International Symposium on Empirical Software Engineering and Measurement* (pp. 181-190). IEEE. <https://doi.org/10.1145/2372251.2372286>

EARLIER EDUCATION

University of Maryland, Baltimore County - MASTER OF SCIENCE (MS), INFORMATION SYSTEMS, MD

2010 - 2012

- **GPA 3.972**
- **Relevant coursework –**

- Advanced Database Projects
- Structural System Analysis and Design
- Project and Systems Engineering Management
- Project Management Operations

- Financial Management
- Advanced Field Research Methods - (Qualitative Research Methods)
- Advanced Experimental Design Methods - (Quantitative Research Methods)

Nagpur University, India BACHELOR OF ENGINEERING (BE), ELECTRICAL ENGINEERING

2002 - 2006